

Lecture: PH of Blood, Acidosis, Alkalosis, Buffer systems

1. At pH 7.4, the ratio of bicarbonate : dissolved CO₂ in plasma is

- a) 1 : 1
- b) 20 : 1
- c) 10 : 1
- d) 40 : 1

2. Quantitatively, the most significant buffer system in plasma is

- (A) Phosphate buffer system
- (B) Carbonic acid-bicarbonate buffer system
- (C) Lactic acid-lactate buffer system
- (D) Protein buffer system

3- The most significant buffer system in kidney is

- (A) Phosphate buffer system
- (B) Carbonic acid-bicarbonate buffer system
- (C) Lactic acid-lactate buffer system
- (D) Protein buffer system

4. A person was admitted in a coma (غيبوبة). Analysis of the arterial blood gave the following values: PCO₂ 16 mm Hg, HCO₃⁻ 5 mmol/l and pH 7.1. What is the underlying acid-base disorder ?

Normal Values: HCO₃ [22 - 28] - PH = 7.35 – 7.45 – P CO₂ [35 – 45]

- a) Metabolic Acidosis
- b) Respiratory Acidosis
- c) Metabolic Alkalosis
- d) Respiratory Alkalosis

5. A student is nervous for a big exam and is breathing rapidly, what do you expect out of the followings

- a) Metabolic Acidosis
- b) Metabolic Alkalosis
- c) Respiratory Acidosis
- d) Respiratory Alkalosis

6. Causes of metabolic alkalosis include all the following except.

- a) Hypokalemia .
- c) Recurrent vomiting

b) Thiazide diuretic therapy.

d) Mineralocorticoid deficiency.

7. Which of the following conditions will not cause respiratory alkalosis?

a) Fever

b) Anxiety

c) Laryngeal obstruction

d) Salicylate toxicity

9-A 45-year-old female with renal failure, missed her dialysis and was feeling sick, what could be the associated condition ?

a) Metabolic Acidosis

b) Metabolic Alkalosis

c) Respiratory Acidosis

d) Respiratory Alkalosis

10-The pH of the body fluids is stabilized by buffer systems. Which one of the following compounds is the most effective buffer system at physiological pH ?

a) Bicarbonate buffer

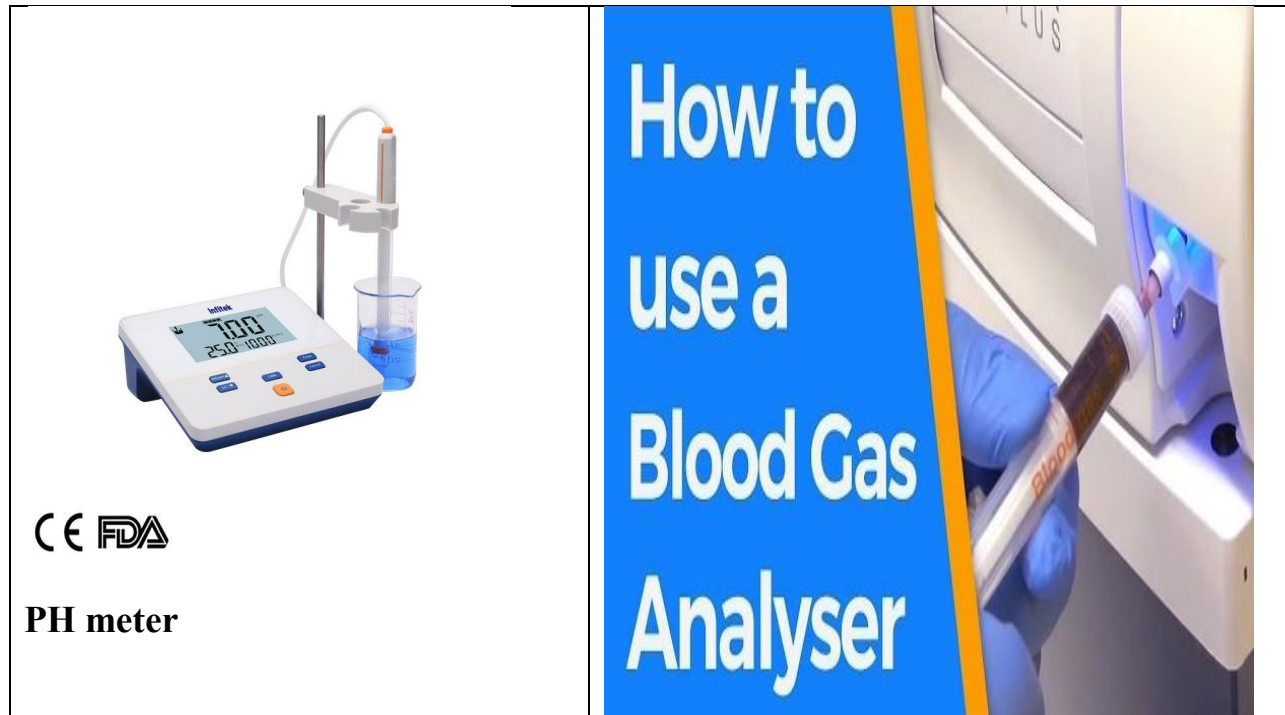
b) Phosphate buffer

c) Protein buffer

d) None of the above

Complete:

1. PH measures:.....hydrogen concentration in the blood
2. The normal pH of the blood is7.35-7.45.....
3. Acidosis of blood means pH is.....below 7.35.....
4. Alkalosis of blood means pH is.....above 7.45.....
5. PH of blood could be measured by.....blood gas analyzer.....
6. PH of laboratory prepared solutions is measured bypH meter.....



True (✓) or False (X) and mention the right answer (Justify):

- a. Acidosis means: high pH is above 7.5 (x). Below 7.35
- b. Acidosis means: PH is below 7. 5 (x). Below 7.35
- c. Normal PH of arterial blood is between 7.35 – 7.45 (✓).
- d. Chronic obstructive lung disease (as Asthma) is complicated with acidosis (✓)..
- e. Chronic cigarette smoking usually leads to respiratory acidosis. (✓).

Lecture: Vitamins and minerals

I- Multiple Choice Questions:

1. Richest source of Vitamin D is

- (A) Cod liver oils (B) Margarine

(C) Egg yolk

(D) Butter and Milk

2. The 7-dehydrocholesterol in skin is cleaved to form cholecalciferol by

(A) Infrared light

(B) Dim light

(C) Ultraviolet irradiation with sunlight.

(D) electrical light

3. The most biologically potent Vitamin D metabolite is

(A) 25-Hydroxycholecalciferol

(B) 1,25-Dihydroxycholecalciferol

(C) 24, 25-Dihydroxycholecalciferol

(D) 1,24 -hydroxycholecalciferol

4. Deficiency of vitamin D in adults causes

(A) Anemia

(C) Hypothyroidism

(B) Tuberculosis of bone

(D) Osteomalacia

5. Which of the following vitamins is fat-soluble?

a- Vitamin A

b- Folic acid

c- Biotin

d- Niacin

6. Which of the followings is water-soluble vitamin?

a) Vitamin A

b) Vitamin D

c) Vitamin C

d) Vitamin K

7. Which of the following statements describe water-soluble vitamins?

a. Their absorption needs bile salts

b. They need carrier protein

c. They can not be stored in the body

d. They include vitamin A&D

8. Which one of the following vitamins is derived from the carotenoid compounds?

- a. Vitamin D
- b. Vitamin B12
- c. Vitamin C
- d. Vitamin A

9.-All the following statements describe rickets Except:

- a. It is caused by deficiency of vitamin D in infants
- b. It is characterized by high plasma calcium and phosphorus levels
- c. It is characterized by box shaped skull
- d. There is knock-knee

10. -All the following vitamins belong to B complex vitamins except:

- a. Riboflavin
- b. Niacin
- c. L. Ascorbic acid
- d. Folic acid

11. FMN and FAD are derived from which one of the following vitamins?

- a- Thiamin
- b- Niacin
- c- Riboflavin
- d- Biotin

12. Which one of the following vitamins enters in the synthesis of coenzyme A?

- a- Folic acid
- b- Nicotinic acid
- c- Pantothenic acid
- d- Ascorbic acid

13. Pernicious anemia is caused by deficiency of one of the following vitamins;

- a- Vitamin B6
- b- Vitamin B1
- c- Vitamin B12 (intrinsic factor)
- d- Folic acid

N.B:- Pernicious anemia is a type of megaloblastic anemia in which the body isn't able to absorb vitamin B12 due to a lack of intrinsic factor in stomach secretions.

14.-Folate deficiency causes

- (A) Microcytic anemia
- (B) Hemolytic anemia
- (C) Iron deficiency anemia
- (D) Megaloblastic anemia

15.-The most important natural antioxidant is

- | | |
|-----------------|---------------|
| (C) Vitamin D | (B) Vitamin E |
| (C) Vitamin B12 | (D) Vitamin K |

16.-Deficiency of vitamin D causes

- (A) Rickets and osteomalacia
- (B) Tuberculosis of bone
- (C) Hypothyroidism
- (D) Skin cancer

17.-Which one of the following statements describe vitamins?

- a- They are required in large quantities for normal growth and health.
- b- They cannot be synthesized by the body in adequate amounts and must be taken in diet.
- c- They enter in the structure of tissues.
- d- They are one of the major sources of energy to the body.

18.-All the following statements describe vitamin C EXCEPT:

- a- It is derived from glucose
- b- 90% of it, present in the form of L-ascorbic acid
- c- it is involved in collagen synthesis
- d- its deficiency causes beriberi

19.-Absorption of Vitamin B12 requires the presence of:

- | | |
|----------------------|-------------|
| (A) Pepsin | (B) Renin |
| (C) Intrinsic factor | (D) amylase |

20.-Deficiency of vitamin A causes

- | | |
|--------------------------|---------------------|
| (A) Osteomalacia | (B) Night blindness |
| (C) Tuberculosis of bone | (D) Cheilosis |

21.-Which of the following vitamin is stored in the liver?

- a. Thiamin
- b. VitaminC
- c. Pantothenic acid
- d. Vitamin A

22.-Which of the following organs is not a target organ for vitamin D action?

- A.Intestine B.Kidney C. Liver D. Bone

23.-Which form of vitamin K is produced by intestinal bacteria?

- a. Phylloquinone(K1)
- b. Menaquinone (K2)
- c. Phyllodione
- d. Menadione (K3)

24.-The synthetic form of vitamin k is:

- a. Phylloquinone(K1)
- b. Menaquinone (K2)
- c. Phyllodione
- d. Menadione (K3)

25.-Which vitamin is important for biosynthesis of clotting factors (II, VII, IX and X)?

- a.Thiamin b.Vitamin C c.Vitamin B12 d.Vitamin K

25- Prolonged prothrombin time (PT time, INR) occurs in case of vitamin.....deficiency.

- a.A b. E c.K d.b12

26- Loosening and falling of teeth, gums is swollen and infected and bleeds easily in vitamin deficiency.

- a.A b.C c.K d.b1

27- The coenzyme derived from vitamin B1 is.....

- a.PLP b.TPP c.NAD d.COASH

28- Biotin serves as a coenzyme forEnzymes.

- a.Dehydrogenases b.Lyases c. Carboxylases d.Ligases

29- About 70% of neural tube defect cases can be prevented by takingprophylactically during pregnancy.

- a. Vitamin D b. Vitamin E **c. Folic acid** d. Pantothenic acid

30- Vitamin D3 synthesis starts in skin fromas a precursor.

- a. Cholecalciferol
b. 7dehydrocholesterol
 c. Ergosterol
 d. Calcitriol

II- Match the following statements in column (A) with the suitable answer in the column (B)

A	The answer	B
1- It is considered as natural antioxidants	(d)	a. Vitamin D
2- Its deficiency causes rickets	(a)	b. Vitamin B12
3- Its coenzyme is FAD & FMN	(e)	c. Vitamin C
4- It needs intrinsic factor for its absorption	(b)	d. Vitamin E
5- It is given prophylactically in pregnancy to prevent spina bifida	(f)	e. Vitamin B2
6- Its deficiency causes scurvy	(c)	f. Folic acid

III- Put true or false and correct the false statement

- 1- Vitamins are organic nutrients which are required in small quantities for normal growth and health (√).
 2- The active form of vit D is 1 hydroxycholecalciferol (x).

1,25 di-hydroxycholecalciferol

3- Eye manifestations of vitamin B12 deficiency is night blindness (x).

vitamin A

4- Vitamin K2 is produced by intestinal bacteria (✓).

5- Deficiency of vitamin E produces scurvy (x). Vitamin C

6-Hypernatremia means an increase in potassium. (x). sodium

7-Hypokalemia means a decrease in potassium.. (✓).

IV- Complete:

1- Thiamine pyrophosphate is the coenzyme of vitamin B1..

2-Pellagra is characterized by: Diarrhia....., Demenia.....,
...Dermatitis.....

3- The coenzymes derived from niacin is ...FMN..... and ...FAD.....

4- Thiamin deficiency causes...Beri Beri.....

5- Ribofalvin is the scientific name of...B2.....

V-Put the suitable answer in front of the following statements:

1- The vitamin that has role in biosynthesis of clotting factors including (2,7,9,10) (Vitamin K).

2- The vitamin used in formation of COASH and Acyl carrier protein (Vitamin B5).

3-The vitamin that provide coenzyme PLP. (Vitamin B6 (Pyridoxine)

4- a glycoprotein secreted by the parietal cells of gastric mucosa and has role in B12 absorption. (Intrinsic factor)

5- Type of anemia due to folate deficiency (megaloblastic anemia)

VII-Short answered questions:

1- Compare between fat & water-soluble vitamins

Comparison between fat & water soluble vitamins

Item	Fat soluble vitamins	Water soluble vitamins
Source	Dietary fat	Water of cooked food
Absorption	Needs bile salts	Don't need bile salts
Carrier protein	Need carrier	Need no carrier
Storage	Can be stored in liver	Not stored in the body
Deficiency	Appear later	Appear early
Urine excretion	Can't be excreted	Easily excreted
Overdose	Leads to hypervitaminosis	Hypervitaminosis is rare

2- Enumerate three manifestations of vitamin A deficiency:

- Night blindness
- Lung infection
- Keratinization of skin

3- List three sources of vitamin K:

Sources:

- ❑ **Vitamin K1**(Phylloquinone): is natural vitamin K found in plants e.g. green leafy vegetables, cabbage, tomatoes and cereals.
- ❑ **Vitamin K2** (Menaquinone) : is produced by intestinal bacteria and found naturally in animal sources e.g. liver and egg yolk
- ❑ **Vitamin K3** (Menadione) : is synthetic vitamin.

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Minerals

Sodium= Natrium (Hyponatremia نقص الصوديوم X Hypernatremia زيادة)

Potassium = Kalium(Hypokalemia نقص البوتاسيوم X Hyperkalemia زيادة)

I- Multiple Choice Questions-

1. All the following minerals are important constituents of bone and teeth except:

- a- Calcium
- b- Phosphorus
- c- Magnesium
- d- Sodium

2. Which one of the following nutritional disorders result from excessive fat accumulation?

- a- Undernutrition
- b- Malnutrition
- c- Obesity
- e- None of the above.

3. Which one of the following diseases result from excessive intake of high salt in Diet?

- a- Obesity
- b- Cancer of the breast
- b- Atherosclerosis
- d- Hypertension

4. - Excessive intake of CHO, fats and salt leads to development of all of the following except:

- a- Obesity
- b- Jaundice
- c. Atherosclerosis
- d- hypertension

5- All of the following are macro-minerals except:

- a-Sodium
- b- potassium
- c- magnesium
- d- zinc

6- All of the following are micro-minerals except:

- a-cobalt
- b- potassium
- c- iron
- d- zinc

Body calcium and phosphorus

1. What percentage of calcium in the human body is stored in bones and teeth?

- A. 50%
- B. 75%
- C. 90%
- D. 99%

☐ **Answer: D. 99%**

2. Which hormone primarily increases blood calcium levels?

- A. Insulin
- B. Parathyroid hormone
- C. Calcitonin
- D. Aldosterone

☐ **Answer: B. Parathyroid hormone**

3. Which of the following is a major function of calcium in the body?

- A. Oxygen transport
- B. Blood clotting
- C. Hormone synthesis
- D. DNA replication

☐ **Answer: B. Blood clotting**

4. Which vitamin is essential for calcium absorption in the intestine?

- A. Vitamin A
- B. Vitamin B12

- C. Vitamin C
- D. Vitamin D

☐ **Answer: D. Vitamin D**

5. Low levels of calcium in the blood (hypocalcemia) can lead to:

- A. Muscle relaxation
- B. Increased bone density
- C. Muscle spasms (tetany)
- D. Decreased nerve activity

☐ **Answer: C. Muscle spasms (tetany)**

6- Deficiency of calcium in adults cause:

- A. Hyperthyroidism
- B. Diabetes mellitus
- C. **Osteomalacia**
- D. Obesity

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2. Approximately what percentage of body phosphorus is found in bones and teeth?

- A. 50%
- B. 70%
- C. 85%
- D. 99%

☐ **Answer: C. 85%**

3. Phosphorus is a key component of a molecule responsible for energy transfer?

- A. DNA
- B. Hemoglobin
- C. ATP
- D. Insulin

☐ **Answer: C. ATP**

4. Which hormone helps regulate phosphate levels by increasing its excretion in the kidneys?

- A. Insulin
- B. Parathyroid hormone
- C. Aldosterone
- D. Calcitonin

☐ **Answer: B. Parathyroid hormone**

5. Which of the following conditions is associated with low phosphate levels?

- A. Hyperphosphatemia
- B. Hypophosphatemia
- C. Hypercalcemia
- D. Hyponatremia

☐ **Answer: B. Hypophosphatemia**

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Body sodium and potassium

1. What is the normal range of sodium in blood plasma?

- A. 120–130 mmol/L
- B. 135–150 mmol/L

- C. 150–160 mmol/L
- D. 160–170 mmol/L

☐ **Answer: B. 135–150 mmol/L**

2. Which hormone plays the most important role in sodium reabsorption in the kidneys?

- A. Insulin
- B. Aldosterone
- C. Parathyroid hormone
- D. Calcitonin

☐ **Answer: B. Aldosterone**

3. Sodium is primarily an important cation in which fluid compartment?

- A. Intracellular fluid
- B. Extracellular fluid
- C. Bone matrix
- D. Mitochondria

☐ **Answer: B. Extracellular fluid**

4. Which condition is caused by low sodium levels in the blood?

- A. Hyperkalemia
- B. Hypernatremia
- C. Hyponatremia
- D. Hypocalcemia

☐ **Answer: C. Hyponatremia**

5. A major function of sodium in the body is:

- A. Formation of hemoglobin
- B. Maintenance of osmotic pressure
- C. Storage of energy
- D. Production of enzymes

☐ **Answer: B. Maintenance of osmotic pressure**

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1. What is the normal range of potassium in blood plasma?

- A. 2.0–3.0 mmol/L
- B. 3.5–5.0 mmol/L
- C. 5.5–7.0 mmol/L
- D. 7.0–9.0 mmol/L

☐ **Answer: B. 3.5–5.0 mmol/L**

2. Potassium is mainly found in which compartment of the body?

- A. Extracellular fluid
- B. Blood plasma
- C. Intracellular fluid
- D. Bone

☐ **Answer: C. Intracellular fluid**

3. Which of the following is a key function of potassium?

- A. Blood clotting
- B. Nerve impulse transmission
- C. Oxygen transport
- D. Bone mineralization

☐ **Answer: B. Nerve impulse transmission**

4. Which condition refers to high potassium levels in the blood?

- A. Hypokalemia
- B. Hypernatremia
- C. Hyperkalemia
- D. Hyponatremia

☐ **Answer: C. Hyperkalemia**

5. Which organ plays the most important role in regulating potassium balance?

- A. Liver
- B. Heart
- C. Kidneys
- D. Lungs

☐ **Answer: C. Kidneys**

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1. What is the normal range of chloride in blood plasma?

- A. 70–85 mmol/L
- B. 90–100 mmol/L
- C. 96–106 mmol/L
- D. 110–125 mmol/L

☐ **Answer: C. 96–106 mmol/L**

2. Chloride is mainly found in which fluid compartment?

- A. Intracellular fluid
- B. Extracellular fluid

- C. Bone
- D. Muscle cells

☐ **Answer: B. Extracellular fluid**

3. Chloride plays an important role in:

- A. Oxygen transport
- B. Acid-base balance
- C. Hormone production
- D. Protein synthesis

☐ **Answer: B. Acid-base balance**

4. Chloride is a major component of which digestive substance?

- A. Bile
- B. Saliva
- C. Gastric acid (HCl)
- D. Pancreatic juice

☐ **Answer: C. Gastric acid (HCl)**

5. Low levels of chloride in the blood are referred to as:

- A. Hyperchloremia
- B. Hypochloremia
- C. Hyponatremia
- D. Hypokalemia

☐ **Answer: B. Hypochloremia**

II- Complete statements:

1. Main intracellular cation is.....K.....
2. Main extracellular cation is.....Na.....
3. Deficiency of potassium is called ...hypokalemia.....
4. Deficiency of sodium is calledhyponatremia.....
5. Body Mass Index (BMI) means...relative body weight to height.....
6. Waist to hip ratio normal in male equals..... ≤ 0.95
7. Normal random blood sugar should be less than ...140 mg/dl.....

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Lecture of Obesity

1. Obesity is defined as having a Body Mass Index (BMI) of:

- A. $\geq 23 \text{ kg/m}^2$
- B. $\geq 25 \text{ kg/m}^2$
- C. $\geq 30 \text{ kg/m}^2$
- D. $\geq 35 \text{ kg/m}^2$

☐ **Answer: C. $\geq 30 \text{ kg/m}^2$**

2. Which of the following is the most common cause of obesity?

- A. Genetic mutation
- B. Hormonal imbalance
- C. Excess caloric intake and low physical activity
- D. Infection

☐ **Answer: C. Excess caloric intake and low physical activity**

3. Which hormone is known as the “satiety hormone” that decreases appetite?

- A. Ghrelin
- B. Leptin
- C. Insulin
- D. Cortisol

☐ **Answer: B. Leptin**

4. Which hormone stimulates hunger?

- A. Leptin
- B. Insulin
- C. Ghrelin
- D. Thyroxine

☐ **Answer: C. Ghrelin**

5. Which of the following conditions is commonly associated with obesity?

- A. Hypoglycemia
- B. Osteoporosis
- C. Type 2 diabetes mellitus
- D. Anemia

☐ **Answer: C. Type 2 diabetes mellitus**

6. Central (abdominal) obesity increases the risk of:

- A. Asthma only
- B. Cardiovascular disease
- C. Skin infections only
- D. Eye disorders

☐ **Answer: B. Cardiovascular disease**

7. Which measurement is most useful to assess central obesity?

- A. Height
- B. Weight
- C. Waist circumference
- D. Arm span

☐ **Answer: C. Waist circumference**

8. Which of the following is a long-term complication of obesity?

- A. Increased immunity
- B. Hypertension
- C. Decreased cholesterol
- D. Improved insulin sensitivity

☐ **Answer: B. Hypertension**

9. Which lifestyle change is most effective for managing obesity?

- A. Increased sugar intake
- B. Sedentary lifestyle
- C. Balanced diet and regular exercise
- D. Skipping meals

☐ **Answer: C. Balanced diet and regular exercise**

10. Obesity increases the risk of which type of joint problem?

- A. Rheumatoid arthritis
- B. Osteoarthritis
- C. Gout only
- D. Bone fractures only

☐ **Answer: B. Osteoarthritis**

11- A person with body height of 180 cm and body weight of 100 kg, what is his BMI?

- A. 23 kg/m²
- B. 27.7 kg/m²
- C. 30 kg/m²
- D. 35 kg/m²

☐ **Answer B: 27.7 kg/m²**

12- A person with a BMI of 26 kg/m² is considered:

- A. Overweight
- B. Obese
- C. normal weight
- D. underweight

☐ **Answer: A. Overweight**

13- A person with a BMI of 17 kg/m² is considered:

- A. Overweight
- B. Obese
- C. normal weight
- D. underweight

☐ **Answer: D. Underweight**